Week 7: SQL Query Implementation for Perfume Shop Management

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Objective

The objective of this project is to implement SQL queries for Perfume Shop Management to efficiently store, retrieve, and manage information about Perfumes, suppliers, customers, sales, and stock details. It aims to use SQL commands to filter and sort records, perform calculations using aggregate functions, monitor medicine availability, analyze sales transactions, and identify perfumes that require restocking, thereby supporting effective daily operations and better decision-making in the Perfume shop.

Perform Basic SQL Queries

SQL SELECT command is used to retrieve data from database tables.

1. Display all customer details.

```
SQL> SELECT * FROM CUSTOMER;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                              PHONE CREATED_A
-------------------------------------------------- ---------- ---------
CITY
------------------------------
C-1001     Chiru                Raju
rajuch06@gmail.com                                 9347855515 17-SEP-26


C-1002     Hema                 Bhaskar
hemabhaskar02@gmail.com                            9884487152 14-JUN-26


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                              PHONE CREATED_A
-------------------------------------------------- ---------- ---------
CITY
------------------------------

C-1003     Seetha               Mani
seethamani16@gmail.com                             9392273469 15-MAY-26


C-1004     Abinaya              Ramesh
abiramesh79@gmail.com                              9710021922 14-SEP-26
```

2. Display all available products.

```
SQL> SELECT * FROM BRAND;

PRODUCT_ID BRAND_NAME                    CATEGORY_I STOCK_QUANTITY      PRICE
---------- ----------------------------- ---------- -------------- ----------
PRODUCT_COUNT
-------------
BR-01      Dior                          CAT-001                25       1500
           1

BR-02      Chanel                        CAT-002                30       1800
           1

BR-03      Gucci                         CAT-003                20       1200
           1


PRODUCT_ID BRAND_NAME                    CATEGORY_I STOCK_QUANTITY      PRICE
---------- ----------------------------- ---------- -------------- ----------
PRODUCT_COUNT
-------------
BR-04      Versace                       CAT-004                15       2200
           1

BR-05      Armani                        CAT-005                35       1400
           1

BR-06      Burberry                      CAT-006                28       1600
           1


PRODUCT_ID BRAND_NAME                    CATEGORY_I STOCK_QUANTITY      PRICE
---------- ----------------------------- ---------- -------------- ----------
PRODUCT_COUNT
-------------
BR-07      Calvin                        CAT-007                10       3500
```

3. Retrieve product names and prices only.

```
SQL> SELECT BRAND_NAME,PRICE FROM BRAND;

BRAND_NAME                          PRICE
------------------------------ ----------
Dior                                 1500
Chanel                               1800
Gucci                                1200
Versace                              2200
Armani                               1400
Burberry                             1600
Calvin                               3500
YSL                                  2500
Fogg                                 2000
Hugo Boss                            1900

10 rows selected.
```

4. Display all orders placed by customers.

```
SQL> SELECT C.FIRST_NAME,C.LAST_NAME, O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

FIRST_NAME           LAST_NAME            ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- ------- ------- --------- ------------
ORDER_STATUS
--------------------
Chiru                Raju                 ORD001  C-1001  25-SEP-26      2500
Delivered

Hema                 Bhaskar              ORD002  C-1002  25-SEP-26      1800
Pending

Seetha               Mani                 ORD003  C-1003  24-SEP-26      3500
Shipped


FIRST_NAME           LAST_NAME            ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- ------- ------- --------- ------------
ORDER_STATUS
--------------------
Abinaya              Ramesh               ORD004  C-1004  23-SEP-26      1200
Delivered

Deepikaa             Selvam               ORD005  C-1005  22-SEP-26      4500
Pending
```

5. Retrieve payment details.

```
SQL> SELECT * FROM PAYMENT;

PAYMENT_ID ORDER_I PAYMENT_D PAYMENT_MODE        PAYMENT_STATUS
---------- ------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY001     ORD001  25-SEP-26 UPI                  Success
              2500

PAY002     ORD002  25-SEP-26 Card                 Pending
              1800

PAY003     ORD003  24-SEP-26 Net Banking          Success
              3500


PAYMENT_ID ORDER_I PAYMENT_D PAYMENT_MODE        PAYMENT_STATUS
---------- ------- --------- -------------------- --------------------
TRANSACTION_AMOUNT
------------------
PAY004     ORD004  23-SEP-26 Cash                 Success
              1200

PAY005     ORD005  22-SEP-26 UPI                  Failed
              4500
```

2. Apply Filtering Conditions Using WHERE

1. Find products with price greater than ₹5000.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE FROM BRAND WHERE PRICE > 5000;

no rows selected
```

2. Display products available in stock.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, STOCK_QUANTITY, CASE WHEN STOCK_QUANTITY > 0 THEN 'IN STOCK' ELSE 'OUT OF STOCK' END AS STOCK_STATUS FROM BRAND;

PRODUCT_ID BRAND_NAME                 STOCK_QUANTITY STOCK_STATUS
---------- ------------------------- -------------- ------------
BR-01      Dior                                  25 IN STOCK
BR-02      Chanel                                30 IN STOCK
BR-03      Gucci                                 20 IN STOCK
BR-04      Versace                               15 IN STOCK
BR-05      Armani                                35 IN STOCK
BR-06      Burberry                              20 IN STOCK
BR-07      Calvin                                10 IN STOCK
BR-09      YSL                                   10 IN STOCK
BR-10      Fogg                                  24 IN STOCK
BR-08      Hugo Boss                             22 IN STOCK

10 rows selected.
```

3. Display completed orders.

```
SQL> SELECT * FROM ORDERS WHERE ORDER_STATUS = 'Delivered';

ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------- ------- --------- ------------ --------------------
ORD001  C-1001  25-SEP-26          2500 Delivered
ORD004  C-1004  23-SEP-26          1200 Delivered
```

4. Find products with rating above 4.

```
SQL> SELECT B.PRODUCT_ID, B.BRAND_NAME, R.RATING FROM BRAND B INNER JOIN REVIEW_RATING R ON B.PRODUCT_ID = R.PRODUCT_ID WHERE R.RATING > 4;

PRODUCT_ID BRAND_NAME                    RATING
---------- ------------------------- ----------
BR-01      Dior                               5
BR-03      Gucci                              5
BR-06      Burberry                           5
BR-09      YSL                                5
```

3. Sort Data Using ORDER BY

1. Display products from lowest to highest price.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE FROM BRAND ORDER BY PRICE ASC;

PRODUCT_ID BRAND_NAME                       PRICE
---------- ------------------------------ ----------
BR-03      Gucci                            1200
BR-05      Armani                           1400
BR-01      Dior                             1500
BR-06      Burberry                         1600
BR-02      Chanel                           1800
BR-08      Hugo Boss                        1900
BR-10      Fogg                             2000
BR-04      Versace                          2200
BR-09      YSL                              2500
BR-07      Calvin                           3500

10 rows selected.
```

1. Display customers alphabetically.

```
SQL> SELECT * FROM CUSTOMER ORDER BY FIRST_NAME ASC;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                          PHONE CREATED_A
---------------------------------------------- ---------- ---------
CITY
------------------------------
C-1004     Abinaya              Ramesh
abiramesh79@gmail.com                        9710021922 14-SEP-26


C-1009     Bhavana              Karthik
bhavanak29@gmail.com                         7338951173 13-SEP-26


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                          PHONE CREATED_A
---------------------------------------------- ---------- ---------
CITY
------------------------------

C-1001     Chiru                Raju
rajuch06@gmail.com                           9347855515 17-SEP-26


C-1005     Deepikaa             Selvam
deepiselva30@gmail.com                       9043707830 15-SEP-26
```

```
CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                          PHONE CREATED_A
---------------------------------------------- ---------- ---------
CITY
------------------------------


C-1002     Hema                 Bhaskar
hemabhaskar02@gmail.com                      9884487152 14-JUN-26


C-1008     Priya                Hari

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL                                          PHONE CREATED_A
---------------------------------------------- ---------- ---------
CITY
------------------------------
haripriya2007@gmail.com                      7339012505 12-AUG-26


C-1006     Sanjana              Doss
sanjanat08@gmail.com                         9003197590 16-SEP-26
```

1. Find top expensive products.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE FROM BRAND ORDER BY PRICE DESC;

PRODUCT_ID BRAND_NAME                      PRICE
---------- ------------------------------ ----------
BR-07      Calvin                           3500
BR-09      YSL                              2500
BR-04      Versace                          2200
BR-10      Fogg                             2000
BR-08      Hugo Boss                        1900
BR-02      Chanel                           1800
BR-06      Burberry                         1600
BR-01      Dior                             1500
BR-05      Armani                           1400
BR-03      Gucci                            1200

10 rows selected.
```

2. Display latest orders first.

```
SQL> SELECT * FROM ORDERS ORDER BY ORDER_DATE DESC;

ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------- ------- --------- ------------ --------------------
ORD001  C-1001  25-SEP-26         2500 Delivered
ORD002  C-1002  25-SEP-26         1800 Pending
ORD003  C-1003  24-SEP-26         3500 Shipped
ORD004  C-1004  23-SEP-26         1200 Delivered
ORD005  C-1005  22-SEP-26         4500 Pending
```

4. Retrieve Unique Values Using DISTINCT

1. Display unique product categories.

```
SQL> SELECT DISTINCT C.CATEGORY_ID, C.CATEGORY_NAME FROM CATEGORY C JOIN BRAND B ON C.CATEGORY_ID = B.CATEGORY_ID;

CATEGORY_I CATEGORY_NAME
---------- ------------------------------
CAT-004    Oriental
CAT-005    Citrus
CAT-009    Aquatic
CAT-003    Fresh
CAT-010    Luxury
CAT-002    Woody
CAT-007    Vanilla
CAT-008    Musky
CAT-001    Floral
CAT-006    Fruity

10 rows selected.
```

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PAYMENT_MODE AS PAYMENT_MODE FROM PAYMENT;

PAYMENT_MODE
--------------------
Net Banking
Card
Cash
UPI
```

5. Search Products Based on Conditions

1. Find products within a specific price range. (Product Between 20 to 50)

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE FROM BRAND WHERE PRICE BETWEEN 20 AND 50;

no rows selected
```

2. Display products belonging to a particular category.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE, CATEGORY_ID FROM BRAND WHERE CATEGORY_ID = '&CATEGORY_ID';
Enter value for category_id: CAT-007
old   1: SELECT PRODUCT_ID, BRAND_NAME, PRICE, CATEGORY_ID FROM BRAND WHERE CATEGORY_ID = '&CATEGORY_ID'
new   1: SELECT PRODUCT_ID, BRAND_NAME, PRICE, CATEGORY_ID FROM BRAND WHERE CATEGORY_ID = 'CAT-007'

PRODUCT_ID BRAND_NAME                      PRICE CATEGORY_I
---------- ------------------------------ ---------- ----------
BR-07      Calvin                           3500 CAT-007
```

1. Find products currently available in inventory.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, STOCK_QUANTITY FROM BRAND WHERE STOCK_QUANTITY > 0;

PRODUCT_ID BRAND_NAME                   STOCK_QUANTITY
---------- ------------------------------ --------------
BR-01      Dior                                   25
BR-02      Chanel                                 30
BR-03      Gucci                                  20
BR-04      Versace                                15
BR-05      Armani                                 35
BR-06      Burberry                               28
BR-07      Calvin                                 10
BR-09      YSL                                    18
BR-10      Fogg                                   24
BR-08      Hugo Boss                              22

10 rows selected.
```

2. Search products using product name.

```
SQL> SELECT * FROM BRAND WHERE BRAND_NAME LIKE '%' || '&SEARCH_NAME' || '%';
Enter value for search_name: Fogg
old   1: SELECT * FROM BRAND WHERE BRAND_NAME LIKE '%' || '&SEARCH_NAME' || '%'
new   1: SELECT * FROM BRAND WHERE BRAND_NAME LIKE '%' || 'Fogg' || '%'

PRODUCT_ID BRAND_NAME                      CATEGORY_I STOCK_QUANTITY      PRICE
---------- ------------------------------ ---------- -------------- ----------
PRODUCT_COUNT
-------------
BR-10      Fogg                           CAT-010                24       2000
            1
```

3. Find low-stock products.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, STOCK_QUANTITY FROM BRAND WHERE STOCK_QUANTITY < 20;

PRODUCT_ID BRAND_NAME                    STOCK_QUANTITY
---------- ------------------------------ --------------
BR-04      Versace                                  15
BR-07      Calvin                                   10
BR-09      YSL                                      18
```

4. Retrieve Customer and Product Information

1. Display customer details with their orders.

```
SQL> SELECT C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME, C.EMAIL, C.CITY, O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                           ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
------------------------------ ------- ------- -------- ------------
ORDER_STATUS
--------------------
C-1001     Chiru                Raju
rajuch86@gmail.com
                               ORD001  C-1001  25-SEP-26         2500
Delivered


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                           ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
------------------------------ ------- ------- -------- ------------
ORDER_STATUS
--------------------
C-1002     Hema                 Bhaskar
hemabhaskar02@gmail.com
                               ORD002  C-1002  25-SEP-26         1800
Pending
```

```
CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                           ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
------------------------------ ------- ------- -------- ------------
ORDER_STATUS
--------------------
C-1003     Seetha               Mani
seethamani16@gmail.com
                               ORD003  C-1003  24-SEP-26         3500
Shipped


CUSTOMER_I FIRST_NAME           LAST_NAME
---------- -------------------- --------------------
EMAIL
--------------------------------------------------
CITY                           ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT
------------------------------ ------- ------- -------- ------------
ORDER_STATUS
--------------------
C-1004     Abinaya              Ramesh
abiramesh79@gmail.com
                               ORD004  C-1004  23-SEP-26         1200
Delivered
```

5. Apply Multiple Filtering Conditions

1. Find perfume products costing more than ₹10000.

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE FROM BRAND WHERE BRAND_NAME LIKE '%Perfume%' AND PRICE > 10000;

no rows selected
```

2. Search products containing the word "Fogg"

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE FROM BRAND WHERE BRAND_NAME LIKE '%Fogg%';

PRODUCT_ID BRAND_NAME                       PRICE
---------- ------------------------------ ----------
BR-10      Fogg                             2000
```

3. Find orders placed within a specific date range

```
SQL> SELECT * FROM ORDERS WHERE ORDER_DATE BETWEEN TO_DATE('&START_DATE','DD-MM-YYYY') AND TO_DATE('&END_DATE','DD-MM-YYYY');
Enter value for start_date: 25-09-2026
Enter value for end_date: 30-09-2026
old   1: SELECT * FROM ORDERS WHERE ORDER_DATE BETWEEN TO_DATE('&START_DATE','DD-MM-YYYY') AND TO_DATE('&END_DATE','DD-MM-YYYY')
new   1: SELECT * FROM ORDERS WHERE ORDER_DATE BETWEEN TO_DATE('25-09-2026','DD-MM-YYYY') AND TO_DATE('30-09-2026','DD-MM-YYYY')

ORDER_I CUSTOME ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------- ------ --------- ------------ ----------------
ORD001  C-1001  25-SEP-26         2500 Delivered
ORD002  C-1002  25-SEP-26         1800 Pending
```

1. Generate Basic Business Reports

Report 1: Product Availability

Report Display:

Product name

Price

Stock quantity

Availability status

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE, STOCK_QUANTITY, CASE WHEN STOCK_QUANTITY > 0 THEN 'AVAILABLE' ELSE 'OUT OF STOCK' END AS AVAILABILITY_STA
TUS FROM BRAND;

PRODUCT_ID BRAND_NAME                       PRICE STOCK_QUANTITY AVAILABILITY
---------- ------------------------------ ------- -------------- ------------
BR-01      Dior                             1500             25 AVAILABLE
BR-02      Chanel                           1800             30 AVAILABLE
BR-03      Gucci                            1200             20 AVAILABLE
BR-04      Versace                          2200             15 AVAILABLE
BR-05      Armani                           1400             35 AVAILABLE
BR-06      Burberry                         1600             28 AVAILABLE
BR-07      Calvin                           3500             10 AVAILABLE
BR-09      YSL                              2500             18 AVAILABLE
BR-10      Fogg                             2000             24 AVAILABLE
BR-08      Hugo Boss                        1900             22 AVAILABLE

10 rows selected.
```

Report 2: Customer Report

Total number of customers

```
     SELECT COUNT(*) AS TOTAL_CUSTOMERS FROM CUSTOMER;
SQL>
TOTAL_CUSTOMERS
---------------
              9
```

New customer registrations

```
SQL> SELECT CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL, PHONE, CREATED_AT FROM CUSTOMER ORDER BY CREATED_AT DESC;

CUSTOMER_I FIRST_NAME          LAST_NAME
---------- -------------------- --------------------
EMAIL                                                    PHONE CREATED_A
-------------------------------------------------- ---------- ---------
C-1001     Chiru                Raju
rajuch06@gmail.com                                   9347855515 17-SEP-26

C-1007     Vanaja               Muniyappan
vanajam2008@gmail.com                                6369040815 17-SEP-26

C-1006     Sanjana              Doss
sanjanat08@gmail.com                                 9003197590 16-SEP-26


CUSTOMER_I FIRST_NAME          LAST_NAME
---------- -------------------- --------------------
EMAIL                                                    PHONE CREATED_A
-------------------------------------------------- ---------- ---------
C-1005     Deepikaa             Selvam
deepiselva30@gmail.com                               9043707830 15-SEP-26

C-1004     Abinaya              Ramesh
abiramesh79@gmail.com                                9710021922 14-SEP-26

C-1009     Bhavana              Karthik
bhavanak29@gmail.com                                 7338951173 13-SEP-26
```

Report 3: Order Report

Total orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS FROM ORDERS;

TOTAL_ORDERS
------------
           5
```

Completed orders

```
SQL> SELECT COUNT(*) AS DELIVERED_ORDERS FROM ORDERS WHERE ORDER_STATUS = 'Delivered';

DELIVERED_ORDERS
----------------
               2
```

Pending orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS FROM ORDERS WHERE ORDER_STATUS = 'Pending';

PENDING_ORDERS
--------------
             2
```

Cancelled orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS FROM ORDERS WHERE ORDER_STATUS = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

Report 4: Product Performance Report

Highest-priced products

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE FROM BRAND ORDER BY PRICE DESC;

PRODUCT_ID BRAND_NAME                         PRICE
---------- ------------------------------ ----------
BR-07      Calvin                               3500
BR-09      YSL                                  2500
BR-04      Versace                              2200
BR-10      Fogg                                 2000
BR-08      Hugo Boss                            1900
BR-02      Chanel                               1800
BR-06      Burberry                             1600
BR-01      Dior                                 1500
BR-05      Armani                               1400
BR-03      Gucci                                1200

10 rows selected.
```

Most reviewed products

```
SQL> SELECT B.PRODUCT_ID, B.BRAND_NAME, COUNT(R.PRODUCT_ID) AS TOTAL_REVIEWS FROM BRAND B INNER JOIN REVIEW_RATING R ON B.PRODUCT_ID = R.PRODUCT_ID
GROUP BY B.PRODUCT_ID, B.BRAND_NAME ORDER BY TOTAL_REVIEWS DESC;

PRODUCT_ID BRAND_NAME                     TOTAL_REVIEWS
---------- ------------------------------ -------------
BR-01      Dior                                       1
BR-04      Versace                                    1
BR-03      Gucci                                      1
BR-07      Calvin                                     1
BR-06      Burberry                                   1
BR-05      Armani                                     1
BR-09      YSL                                        1
BR-10      Fogg                                       1
BR-02      Chanel                                     1
BR-08      Hugo Boss                                  1

10 rows selected.
```

Available products

```
SQL> SELECT PRODUCT_ID, BRAND_NAME, PRICE, STOCK_QUANTITY FROM BRAND WHERE STOCK_QUANTITY > 0;

PRODUCT_ID BRAND_NAME                         PRICE STOCK_QUANTITY
---------- ------------------------------ ---------- --------------
BR-01      Dior                                 1500             25
BR-02      Chanel                               1800             30
BR-03      Gucci                                1200             20
BR-04      Versace                              2200             15
BR-05      Armani                               1400             35
BR-06      Burberry                             1600             28
BR-07      Calvin                               3500             10
BR-09      YSL                                  2500             18
BR-10      Fogg                                 2000             24
BR-08      Hugo Boss                            1900             22

10 rows selected.
```